

Spatial Cluster Randomized Trials - Sampling Design with Spillover Effects & Spatial Dependence

Andrew Walther

July 28, 2025

Abstract

Background: Investigators designing cluster randomized trials desire insights directing treatment assignment methodology for studies in the presence of spillover effects and spatial dependence.

Methods: Treatment assignment methods including random sampling and block stratified are applied and spatial autoregressive modeling is applied with consideration for spillover effects and spatial dependence for estimation of intervention effects.

Simulation: Simple Random sampling & Block Stratified Sampling treatment assignment methods were applied to spatial grids with dimensions of 2x4, 3x3, and 3x4 clusters (cells). Varieties of spillover effects and spatial dependence sampled observations were considered for estimation of the intervention effect via a spatial autoregressive (SAR) model.

Results: Findings of an extensive simulation study considering the comparison of block stratified and random treatment assignment sampling methods with consideration of spillover effects, spatial dependence, and regional spillover restrictions indicate that randomly selected treatment assignments result in reduced average error when estimating intervention effects, but block stratified treatment assignments lead to reduced variation among a series of estimates. This is an indication a block stratified treatment arrangement won't achieve the minimal level of estimation error, but it remains consistent across a range of selected parameters.

Conclusion: A clear relationship between spillover effect magnitude and MSE for intervention effect estimation is apparent. MSE for the intervention effect is minimized for random sampling treatment assignment, but block stratified assignment minimizes error variance.

1 Introduction

1.1 Background & Motivation

Randomized controlled trials (RCTs) are a cornerstone of evidence-based research, but logistical constraints or the nature of an intervention often necessitate randomizing groups of individuals in a cluster randomized trial (CRT)[12][15]. CRTs are particularly useful for evaluating public health interventions where treatment contamination is a concern and for estimating spillover effects which are the indirect effects of an intervention on untreated individuals neighboring areas that received an intervention[15][17].

When clusters are defined by geography, the presence of spillover effects introduces significant analytical challenges by violating the assumption of cluster independence[1][19]. These challenges are compounded by spatial dependence which is the principle that nearby observations tend to be more related than distant ones[19][29]. The presence of spatial dependence can further bias estimates of intervention efficacy.

Despite the common use of geographical units as clusters, the explicit consideration of these spatial effects at the design stage is rare, and a standard methodology for addressing them hasn't been proposed[19]. A systematic review of existing literature suggests that existing spatial CRTs rarely discuss the method of treatment allocation as a strategic choice to mitigate these complexities. Rather, spatial effects have been left as an afterthought when considering treatment sampling methodology in

spatial CRTs[19]. This results in a lack of simple, intuitive recommendations for investigators, creating a critical need for research into sampling design strategies that can effectively account for both spillover effects and spatial dependence to yield unbiased and efficient estimates of intervention effects.

1.2 Aims & Contributions

This study aims to investigate the efficiency of treatment assignment sampling methodologies in spatial cluster randomized trials for estimating the intervention effect in the presence of both spatial spillover effects with consideration for spatial dependence. We compare two different sampling approaches for treatment assignment on clusters: Simple Random Sampling (SRS) and Block Stratified Sampling (BSS). BSS specifically seeks to isolate clusters assigned as intervention or control from other clusters of the same treatment assignment. Our simulation study will explore how estimation error of the intervention effect in a Spatial CRT varies across a range of spillover effect sizes, the presence and absence of spatial dependence, and varying restrictions on clusters where spillover is permitted to occur. A strong understanding of patterns that exist across these sampling methods and other features considered provides practical guidance to investigators designing Spatial CRTs, assisting them with selecting sampling strategies that align with their goals to minimize bias and/or variance of intervention effect estimates in a spatial setting.

1.3 Overview

The remainder of this article is organized as follows: Section 2 outlines the theoretical framework of the study, including the data generation process, sampling procedures, and estimation methods employed. Section 3 details algorithm and results of the simulation study undertaken to evaluate the sampling methods considered. Section 4 considers a real-world application of these findings to make a recommendation for other investigators. Finally, Section 5 discusses the implications of our findings, potential drawbacks to the work performed, and potential areas for continued future research.

2 Methods

2.1 Cluster Randomized Trials

The randomized controlled trial (RCT) serves as the gold standard design to evaluate intervention efficacy (Kim et al., 2021). Its core principle, the random allocation of individual participants, is expected to balance both measured and unmeasured confounding variables across treatment arms[20]. However, individual randomization is often impractical, or an intervention is delivered at a group level, necessitating an alternative approach[12][15]. In these scenarios, the cluster randomized trial (CRT) is employed, where intact social units or groups like households, clinics, or geographic areas are randomized to intervention arms[12][24]. A key advantage of this design is the prevention of treatment contamination, a scenario where control group members are inadvertently exposed to the intervention, which would otherwise bias the estimated treatment effect toward the null[15]. Statistical considerations in a CRT are the measured response from an intervention and the inherent correlation among outcomes of individuals within the same cluster, a dependency quantified by the intraclass correlation coefficient (ICC)[12][16]. The ICC measures how similar the outcomes of individuals within a cluster are likely to be, relative to those of other clusters. This intra-cluster correlation violates the standard assumption of independence between observations. The failure to account for this feature in the analysis leads to underestimated standard errors and an inflated probability of a Type I error[11][25]. Therefore, the valid design and analysis of CRTs require specialized statistical methods that properly account for this clustering structure, with comprehensive frameworks provided by Donner and Klar (2000)[12] and Hayes and Moulton (2017)[15].

2.2 Neighbor Definitions

2.2.1 Contiguity-Based

Contiguity-based neighbors[13] are constructed with the assumption that neighbors of a given area (individual cells in the case of this study) are other clusters (or cells) that share a common boundary. This boundary may be an edge and/or vertex. The **spdep** R package[6][10][28][8] includes functions for dealing with spatial neighbors and spatial dependence structures. In this study, we consider Rook Neighbors, which fulfill a contiguity condition where a pair of clusters share a common boundary made up by an edge. Another type of contiguity-based neighbors are Queen neighbors that fulfill a contiguity condition where a pair of clusters share a common boundary of either an edge and/or a vertex[13]. Clusters may also have Bishop neighbors where a common boundary at the vertex (corner) of the spatial region is shared. Clusters may also have higher-order neighbors where they do not directly share an edge and/or vertex, but rather have a common boundary with a cluster that is a neighbor of another cluster. In other words, these are “neighbors of neighbors” of the reference cluster.

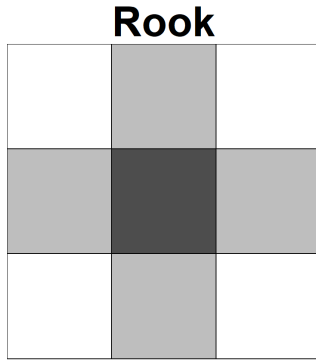


Figure 1: Rook neighbors (light) share an edge with a reference spatial region (dark)[21]

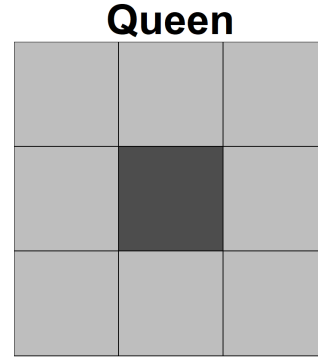


Figure 2: Queen neighbors (light) share either an edge or vertex (or both) with a reference spatial region (dark)[21]

Figure 3: Comparison of Rook and Queen Contiguity Definitions

2.2.2 Distance-Based

Distance-based neighbors[13] are constructed with the assumption that neighbors of a given area (individual cells in the case of this study) are other areas (or cells) that lie within a specified distance of another cell or observation. This distance is commonly delineated by the radius of a circle around the center of a reference spatial region. The region specified in Figure 4 identifies distance-based neighbors of the reference cell (dark) as neighboring cells where the centroid of the cells lie within the circumscribed boundary around the reference cell. In addition to cells, distance-based neighborhoods can also be constructed based on individual points that are within a specified distance of another point.

2.3 Intervention & Spillover Effects

Two primary effect types are considered to understand how treatments affect subjects where they are applied and other subjects in the surrounding areas. The intervention effect and spillover effect are two parameters that are estimated in this study. Spatial cluster randomized trials have subjects or regions that are assigned to receive an intervention and others who are assigned as controls that do not receive an intervention. Observation of the downstream outcomes allow for estimation of the direct intervention effect and indirect spillover effect on the study region.

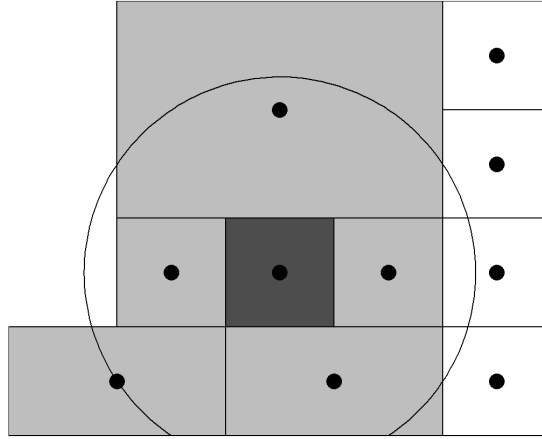


Figure 4: Distance-based neighbors (light) have centroids within a fixed radius of the reference cluster (dark). Clusters with centroids beyond this distance are not considered as neighbors (white)[21]

2.3.1 Intervention Effect

The intervention effect (β) represents the estimated effect of the direct intervention applied in a study. This is a measure of the study outcome that is directly attributed to the intervention that is assigned and applied in the study. Estimation of this intervention effect is possible due to the presence of subjects or regions assigned as a control (that do not receive an intervention) for comparison. Over an array of applications the intervention effect could be a vaccine to prevent infectious disease, a training program for government employees, an educational program for students, and many more.

2.3.2 Spillover Effect Methods

The spillover effect (ψ) represents an indirect effect observed from the intervention on subjects or regions where an intervention was not directly applied[3][4]. There are a number of ways to model a spillover effect as it is observed due to spatial proximity to subjects or regions where an intervention is applied. In this study, only binary spillover effects were considered. A binary spillover effect is a fixed level of a spillover effect that is applied if a subject or region is deemed to be a neighbor of a subject or region where the intervention is applied. The response for subject i considering this spillover effect is expressed as

$$y_i = \alpha + \beta x_i + \psi z_i + \varepsilon_i. \quad (1)$$

where x_i is an indicator of whether subject i receives the intervention, z_i is an indicator of whether the same subject neighbors another subject that receives the intervention (1) or not (0) and is eligible to experience spillover, and ψ is the maximum proportion of the intervention effect that can be experienced as spillover - in this case the binary effect applies 100% of possible spillover. Alternative methods of modeling a spillover effect include: linear, categorical, exponential, and gaussian decay functions[18][5].

2.3.3 Spillover Effect Application

In this study, there are two specific ways in which spillover of the applied intervention is allowed to propagate to other cells or subjects. In both cases, spillover effects originate from areas where the original intervention treatment is applied. The first case allows for spillover to propagate from cells assigned as receiving the intervention into only cells assigned as control cells. In this case, the effect of the intervention is not permitted to spillover into other intervention cells. This suggests a case where the intervention effect applied is the maximum observable effect and any potential spillover cannot increase the observed effect of a particular intervention. Another method of spillover application is

considered where spillover effects are permitted to be applied to both control AND intervention cells that are neighbors of a cell that receives an intervention. This application suggests that an effect greater than that of the original intervention applied can be potentially be observed in a cell that received an intervention and also had spillover from another intervention cell. It is also to be expected that any change in the estimated effects would not be observed in spatial settings where the treatment allocation does not satisfy block stratification conditions (where no intervention cells are rook neighbors with other intervention cells). A representative comparison of these two methods of permissibility of spillover effects is given in Figure 5. In all scenarios of this simulation study, the application of any spillover effect is simply a binary event of whether individuals in a cluster receive a spillover effect or not. Even if spillover exposure could potentially be directed from multiple neighboring clusters, the modeling in this study does not consider any additive spillover effects.

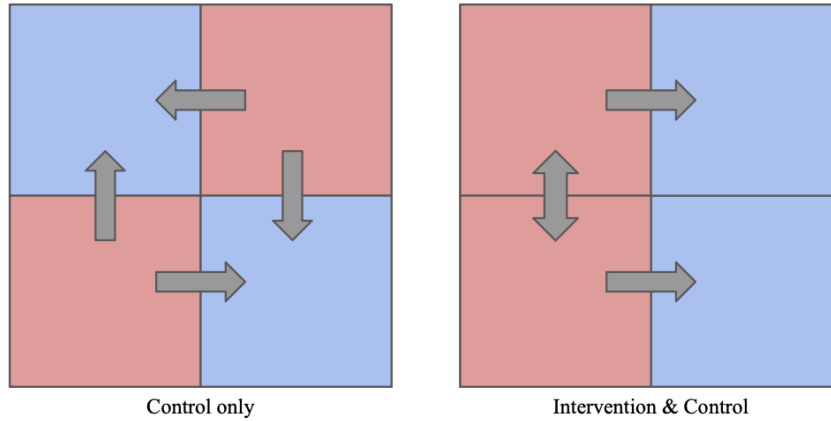


Figure 5: (Left) Spillover application from intervention cells (red) to only control cells (blue) & (Right) spillover application to intervention (red) and control cells (blue).

2.4 Sampling Methods - Simple Random & Block Stratification

In this study, two different treatments are considered: intervention & control. Assignment of these treatments is considered under two methods: Simple Random Sampling (SRS) & Block Stratified Sampling (BSS). Random sampling treatment assignment considers all possible combinations of intervention & control arrangements on a specified spatial grid and provides equal probability of assignment for any given combination of intervention and control. For example, a 2 row by 3 column spatial grid has 6 total cells. Consideration of a 1:1 ratio between intervention and control treatments provides that three cells are assigned as intervention and three cells are assigned as control. Therefore, there are $\binom{6}{3} = 20$ different combinations of treatments for this specific spatial region.

Block stratified sampling seeks to ensure that for a given balanced number of intervention and control cells, that no intervention or control cells share a common edge boundary with each other. An exception to this rule occurs in spatial setups where both dimensions are odd (ie. 3x3) such that the count of intervention and control cells must be out of balance and we assert that the control class will always be in the majority. In these instances, we relax the common edge boundary restriction for control cells and only require that intervention cells not share a common edge boundary with each other in order to satisfy the block stratification condition.

We now consider a series of spatial grid setups to address efficient estimation of the intervention effect. These spatial grids are denoted as (row $\times$ column) in terms of dimension and we seek to apply a balanced distribution between intervention and control cells where possible. When this isn't possible, we defer to the control group being the majority class (more cells) as mentioned previously.

2.4.1 2x4 Grid

One representative spatial setting considered is grid of 8 individual cells laid out in an arrangement of 2 rows and 4 columns. For this grid setup, a balanced set of intervention and control cells can be assigned where 4 cells are assigned as intervention regions and 4 cells are assigned as control regions. For a grid of 8 total cells and 4 cells of each treatment type, there are $\binom{8}{4} = 70$ unique arrangements that cells can be assigned between treatment and control. Figure 6 (left) displays one of these combinations that satisfies the block stratified condition where intervention and control cells do not have any rook neighbors of the same treatment type. For this spatial setting, there are 2 (out of 70) combinations that satisfy the block stratification condition. Figure 6 (right) displays an example of the 2x4 grid spatial setting with a randomly sampled arrangement of intervention and control cells. Note that both treatment types have cells that are rook neighbors with the same treatment. These spatial grids are constructed with Simple Features for R package[27][28].

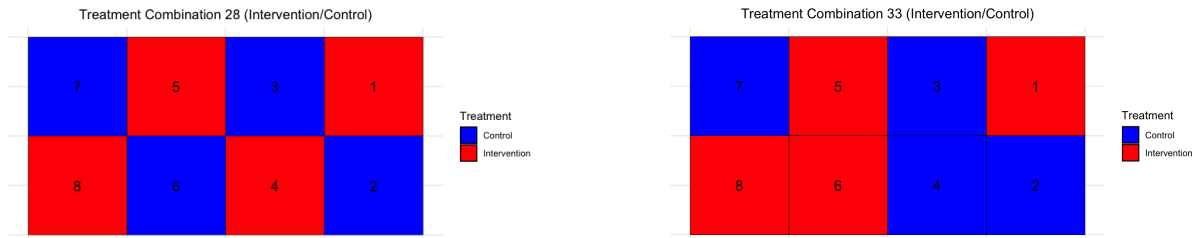


Figure 6: Comparison of intervention (red) and control (blue) clusters assigned via Block Stratified Sampling (left) and Simple Random Sampling (right) on a 2x4 Grid

2.4.2 3x3 Grid

The 3x3 symmetric grid setting of 9 cells allows for $\binom{9}{4} = \binom{9}{5} = 126$ unique combinations of intervention and control cells where 5 cells are allocated as control and 4 cells are assigned as treatment. The majority class of 5 cells is deemed to be the control for the sake of hypothetical resource management. Figure 7 (left) illustrates one combination of treatment and control cells that completely satisfies the block stratification conditions. Additional cluster combinations satisfy a relaxed condition of block stratification where control cells are allowed to have other control cells as rook neighbors. This is possible in spatial grid settings where both dimensions (row, column) are odd in length. There is one combination for the 3x3 grid setting that satisfies the strongest block stratification condition and 5 more combinations that satisfy the relaxed block stratification condition for a total of 6. An example of randomly sampled treatment assignment that doesn't satisfy block stratification conditions is provided in Figure 7 (right).

2.4.3 3x4 Grid

The final spatial setting considered was a 3 row by 4 column (3x4) grid of 12 total cells. This grid, with a balanced allocation of 6 intervention cells and 6 control cells allows for $\binom{12}{6} = 924$ unique arrangements of the two treatments in the spatial region. For this particular grid size, 2 of the 924 treatment combinations satisfy the block stratification condition and Figure 8 (left) presents one of these, with a “checkerboard” design. The remaining treatment combinations are achieved via random sampling of a balanced number of intervention and control cells. Figure 8 (right) illustrates another randomly sampled arrangement of the treatments onto the spatial area where control and intervention cells each have rook neighbors of the same treatment type.

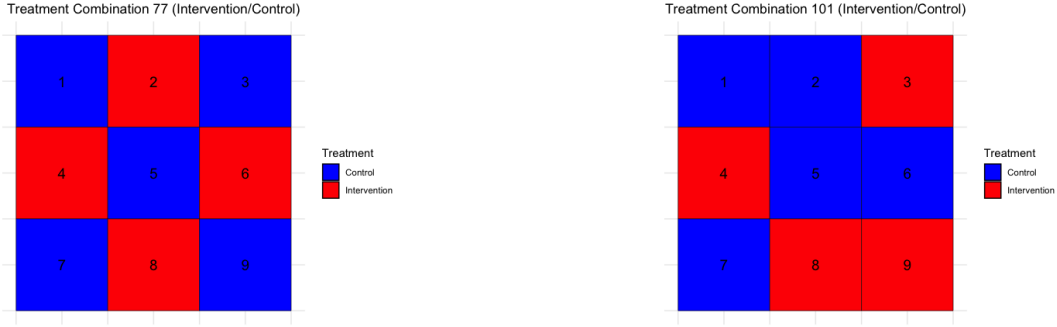


Figure 7: Comparison of intervention (red) and control (blue) clusters assigned via Block Stratified Sampling (left) and Simple Random Sampling (right) on a 3x3 Grid

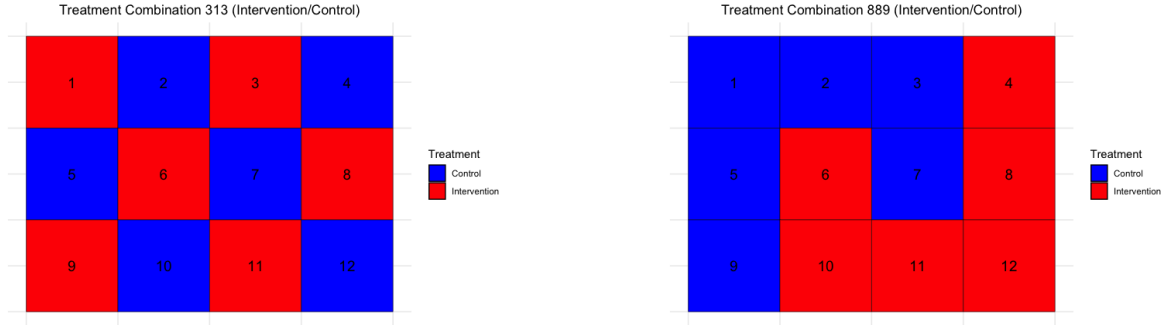


Figure 8: Comparison of intervention (red) and control (blue) clusters assigned via Block Stratified Sampling (left) and Simple Random Sampling (right) on a 3x4 Grid

2.5 Estimation - Spatial Autoregressive (SAR) Model

Spatial dependence of observations can be considered via modeling with a spatial autoregressive model[31][13][30]. A spatial autoregressive model has the form:

$$y_i = \rho \sum_j w_{ij} y_j + \mathbf{x}_i \beta + \varepsilon_i \quad (2)$$

where ρ is the spatial correlation (autoregressive) coefficient and ε_i represents an i.i.d. error term. This can also be expressed in matrix notation as:

$$\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \mathbf{X} \beta + \varepsilon \quad (3)$$

It follows that the general form of the response for this model for the purpose of this study is

$$y_i = \alpha + \beta x_i + \psi(d_i) z_i + \rho \mathbf{W} \mathbf{y} + \epsilon_i \quad (4)$$

where y_i is the observed response for subject i , α is the baseline/background effect, β is the intervention effect, x_i is an indicator of whether subject i receives the intervention, $\psi(d_i)$ is the spillover effect expressed as a function of distance from the nearest intervention source for subject i , z_i is an indicator of whether subject i can experience any spillover effect, and $\rho \mathbf{W} \mathbf{y}$ captures spatial dependence from neighboring observations and their effects in the response variable. The components of $\rho \mathbf{W} \mathbf{y}$ are ρ , a fixed spatial correlation parameter, $\mathbf{W}$, the spatial weights matrix indicating the strength of relationship between neighboring observations, and the response vector $\mathbf{y}$. Finally, ε_i is a random error term for each subject.

Parameter estimates from the spatial lag model are obtained via maximum likelihood estimation[26][9][7]. When considering spatial dependence, the responses are

$$y_i = \rho \sum_{j \neq i} w_{ij} y_j + \alpha + \beta x_i + \psi z_i + \epsilon_i \quad (5)$$

where the residuals are

$$\epsilon_i = y_i - \rho \sum_{j \neq i} w_{ij} y_j - (\alpha + \beta x_i + \psi z_i). \quad (6)$$

Then, assuming that $\epsilon_i \sim N(0, \sigma^2)$, the likelihood function is

$$L(\alpha, \beta, \psi, \rho, \sigma^2) = \frac{1}{\sqrt{(2\pi\sigma^2)^n |\mathbf{I} - \rho\mathbf{W}|}} \exp\left(-\frac{1}{2\sigma^2} \epsilon^\top \epsilon\right) \quad (7)$$

where $|\mathbf{I} - \rho\mathbf{W}|$ is the determinant of $(\mathbf{I} - \rho\mathbf{W})$. The log-likelihood is

$$\log L(\alpha, \beta, \psi, \rho, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) + \log |\mathbf{I} - \rho\mathbf{W}| - \frac{1}{2\sigma^2} \sum_{i=1}^n \left(y_i - \rho \sum_{j \neq i} w_{ij} y_j - (\alpha + \beta x_i + \psi z_i) \right)^2 \quad (8)$$

which is maximized with respect to $\alpha, \beta, \psi, \rho$, and σ^2 . In applied simulation tasks, estimates can be obtained via the ‘lagsarlm()’ function from the R spdep library[6].

2.6 Spatial Dependence

Spatial dependence, used interchangeably with spatial autocorrelation, refers to the property of spatial data where observations close to each other tend not be independent of each other. Rather, observations that are close to each other tend to be dependent on each other and influence values of surrounding data. Hence, there is an implied systematic correlation between data at neighboring locations. Spatial correlation can be positive, where similar values tend to be clustered close together, or negative, where dissimilar values are close together. Appropriate consideration of spatial dependence is necessary for performing valid statistical inference as well as for accurate estimation of effects that are modeled. A pair of common methods for assessing spatial autocorrelation are Moran’s I [22][23] and Geary’s C [14][31]. Moran’s I standardizes the spatial auto-covariance by the variance of the data and Geary’s C utilizes the sum of the squared differences between pairs of data as a measure of co-variation. These test statistics are reliant on the spatial weights matrix discussed in Section 2.6.1.

2.6.1 Spatial Weights Matrix

The spatial weights matrix $\mathbf{W}$ is a component of the SAR model from Section 2.5. It is composed elements w_{ij} which indicate the strength or presence of a spatial connection between observations i and j . Therefore the spatial dependence term of the SAR model is represented as

$$\rho \mathbf{W} \mathbf{y} = \rho \sum_{j \in \mathcal{N}(i)} w_{ij} y_j \quad (9)$$

where ρ is the spatial autocorrelation parameter; an indicator of how strongly neighboring values affect the focal observation, $\mathbf{W}$ is the spatial weights matrix between subjects i and j with matrix elements w_{ij} that are either indicator-based ($\{0, 1\}$) or distance-based $\left(\frac{1}{d_{ij}}\right)$ where d_{ij} is the distance between subjects i and j , and $\mathcal{N}(i)$ is the set of all neighbors of observation i . In this study, this set consists of all subjects that are a member of a cluster that is a rook neighbor of the cluster of subject i . For the spatial weights matrix, The indicator-based weights are

$$w_{ij} = \begin{cases} 1, & \text{if points } i \text{ and } j \text{ are neighbors} \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

and are normalized via

$$W_{ij} = \frac{w_{ij}}{\sum_j w_{ij}} \quad (11)$$

and the distance-based weights are

$$w_{ij} = \begin{cases} \frac{1}{d_{ij}}, & \text{if } 0 < d_{ij} < d_{\max} \\ 0, & \text{otherwise} \end{cases} \quad (12)$$

which are normalized via

$$W_{ij} = \frac{\frac{1}{d_{ij}}}{\sum_j \frac{1}{d_{ij}}}. \quad (13)$$

3 Simulation Study & Results

A simulation study was carried out to consider the contrast of block stratified and random treatment assignment sampling methods with consideration for spillover effects, spatial dependence, and restrictions on spillover application between intervention and control regions. The results of this study are detailed in the remainder of this section.

3.1 Response Generation

Given that the form of the spatial lag model contains the response on both sides of the expression, a separate closed-form expression for data generation must be considered[13][2]. This data generation expression is

$$\mathbf{y} = (\mathbf{I} - \rho \mathbf{W})^{-1} (\boldsymbol{\alpha} + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) \quad (14)$$

where $X\boldsymbol{\beta} = \beta x_i + \psi z_i$. x_i is an indicator of whether subject i receives the intervention and z_i is an indicator of whether subject i receives any spillover effect.

3.2 Simulation Scenarios

We carry out a simulation study to explore how estimation error of the intervention effect in a Spatial CRT varies across a range of spillover effect sizes, the presence or absence of spatial dependence, varying restrictions on clusters where spillover is permitted to occur, and the method of sampling for treatment assignment. Each simulation scenario is composed of a set of these parameters. The baseline effect, α , is always assigned as 0.2. The intervention effect, β , is always assigned as 1.0. The spillover effect, ψ , has possible values 0.5, 0.6, 0.7, and 0.8. The spatial dependence parameter, ρ is set as either 0.00 or 0.01. The sampling design for each scenario is either Block Stratified Sampling (BSS) or Simple Random Sampling (SRS). Finally, the spillover application type is (1) Control Clusters Only or (2) Control & Intervention Clusters. Each simulation setting is replicated for each of the spatial grid dimensions discussed previously (2x4, 3x3, and 3x4).

3.3 Simulation Algorithm

The algorithm implemented to perform the simulations for this study is illustrated in the following section. The simulation is performed over all potential cluster treatment combinations (intervention/control) for each of the 2x4, 3x3, and 3x4 cluster spatial grid geometries. The simulation procedure is as follows:

1. Define the spatial grid geometry & set of parameters for simulation scenario (α , β , ψ , ρ)
2. List out all cluster treatment combinations for specified spatial grid (70 combinations for 2x4, 126 combinations for 3x3, 924 combinations for 3x4)

3. Construct the spatial grid object for the corresponding number of clusters (8, 9, 12) and define the rook neighbors for each cluster in the spatial region.
4. Identify cluster treatment combinations for the specified spatial grid object that satisfy Block Stratified Sampling (BSS) conditions (2x4: 2, 3x3: 6, 3x4: 2).
5. For each combination in the set of all cluster treatment combinations for a given spatial grid:
 - (a) Sample 20 subjects per cluster over the entire space of the grid (2x4: 160 subjects, 3x3: 180 subjects, 3x4: 240 subjects).
 - (b) Assign each subject to its corresponding cluster and to the intervention or control treatment based on the treatment assignment of the specific cluster.
 - (c) Identify the rook neighbors of each subject and whether each subject is exposed to a binary spillover effect based on the treatment assignment of neighboring clusters and selected spillover application type as described in Section 2.3.3.
 - (d) Generate the response value for each subject with consideration for spatial dependence via Spatial Autoregressive (SAR) modeling.
 - (e) Estimate and retain parameter values for α, β, ψ , and ρ with the Spatial simultaneous autoregressive lag model ('lagsarlm()') via maximum likelihood estimation.
 - (f) Iterate parts (a)-(e) of Step 5 N=50 times for each cluster treatment combination for the 2x4 and 3x3 spatial grids and N=10 times for each cluster treatment combination for the 3x4 spatial grid.
6. Aggregate parameter estimates for each iteration of each cluster treatment combination for a specified spatial grid
7. For each parameter estimate ($\alpha, \beta, \psi, \rho$), compute the bias of the mean of the estimates relative to the true value, variance of the estimates, and Mean Squared Error (MSE) of the estimates such that these metrics are computed for each cluster treatment combination.
8. Repeat Steps 1-5 of the simulation procedure for each spatial grid setup (2x4, 3x3, 3x4).

Figure 9 presents an example of the 8 cluster 2x4 spatial grid object. 160 subjects (20 subjects for each cluster) with uniformly distributed coordinates are sampled over the area of the spatial object and colored according to their cluster membership. Subject counts are not necessarily identical between any pair of clusters as the sample of subjects is considered over the whole space. Points in any cluster on the grid are subject to the selected treatment (intervention/control) and equally experience any potential spillover along with all other points that are members of that same cluster. Similar illustrative figures of the sampled observations over the spatial grid could be constructed and displayed for the other spatial geometries under consideration in the simulation study.

3.4 Simulation Evaluation Metrics

Estimates for the baseline effect (α), intervention effect (β), spillover effect (ψ), and spatial correlation parameter (ρ) were collected for each iteration of the simulation procedure for every treatment assignment combination. Considering the initial fixed simulation parameters, the following metrics are computed for estimation of the intervention effect for the evaluation of each set of simulation conditions.

- **Bias:** a measure of how different the estimated parameter value is from the true parameter value. Small bias means that the parameter estimate is quite close to the true value.
- **Variance:** a measure of the spread of a set of estimates from multiple simulation iterations. Smaller variance indicates that the estimates are tightly grouped together and large variance means that the estimates are spread out

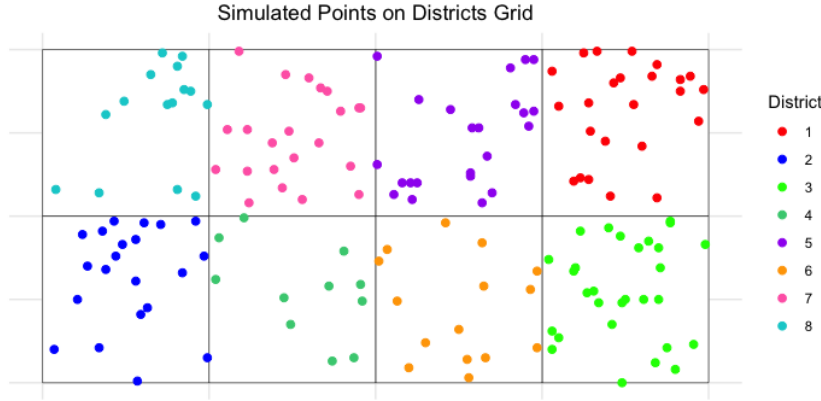


Figure 9: Example of simulated observation coordinates ($n=160$) on a 2×4 grid of clusters

- **Mean Squared Error (MSE):** measures the average squared difference between each estimate and the true value of a parameter. Mean Squared Error combines variance to understand the consistency of a set of estimates as well as bias to understand any trend in deviation from the true value into a measure of estimation accuracy.

Mean Squared Error is the primary metric of consideration to understand the efficient estimation of the intervention effect under all simulation settings. The following subsections detail the findings from the simulation study, focusing on the Mean Squared Error (MSE), standard deviation (SD), and bias of the intervention effect (β) estimates under various conditions (simulation scenarios). The analysis compares Simple Random Sampling (SRS) and Block Stratified Sampling (BSS) across different spatial grid configurations, spillover effect magnitudes, spatial dependence levels, and spillover application types.

3.5 Intervention Effect Estimates (2×4 grid)

For the 2×4 spatial grid, the simulation results for the intervention effect estimates are presented in Table 1 and visually represented in Figure 10.

When spillover was applied to Control Clusters Only (TrtNoSpill), SRS generally exhibited higher MSE compared to BSS, particularly as the spillover effect (ψ) increased. For instance, with $\rho = 0.00$ (no spatial dependence), SRS MSE ranged from 0.17278 ($\psi = 0.5$) to 0.44075 ($\psi = 0.8$), while BSS maintained very low MSE values (e.g., 0.00790 for $\psi = 0.5$). Similarly, with $\rho = 0.01$ (spatial dependence), SRS MSE increased from 0.30384 ($\psi = 0.5$) to 0.57345 ($\psi = 0.8$), whereas BSS showed MSE values ranging from 0.16515 to 0.56900. BSS consistently demonstrated smaller standard deviations compared to SRS under this spillover application type, indicating greater precision. SRS estimates generally showed negative bias, which increased in magnitude with higher ψ .

When spillover was applied to Intervention & Control Clusters (TrtSpill), SRS generally showed lower MSE compared to BSS, especially in the absence of spatial dependence ($\rho = 0.00$). For $\rho = 0.00$, SRS MSE ranged from 0.00790 ($\psi = 0.5$) to 0.01872 ($\psi = 0.8$), while BSS MSE was consistently higher (e.g., 0.26160 for $\psi = 0.5$). With spatial dependence ($\rho = 0.01$), SRS still maintained lower MSE (0.09543 to 0.16216) compared to BSS (0.16515 to 0.56900). SRS also exhibited smaller standard deviations and bias magnitudes under this condition, suggesting better overall accuracy and precision.

Across both spillover application types for the 2×4 grid, the presence of spatial dependence ($\rho = 0.01$) consistently led to higher MSE values for both sampling methods compared to scenarios without spatial dependence ($\rho = 0.00$).

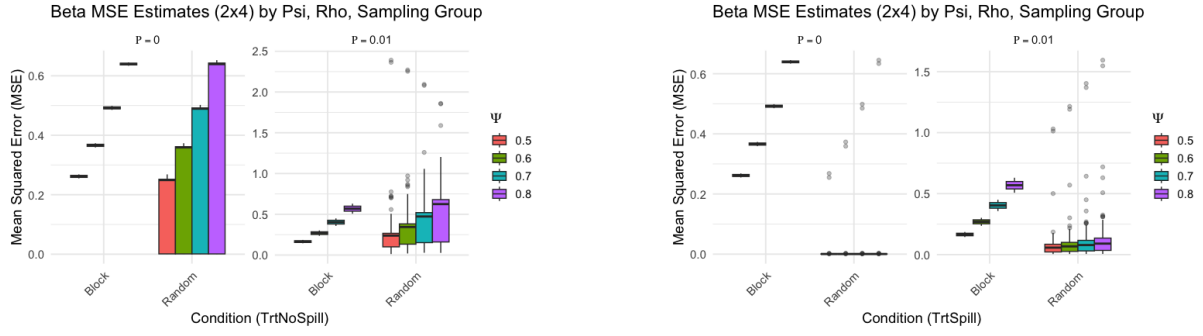


Figure 10: Intervention effect MSE for 2x4 cluster grid with spillover allowed on control and intervention clusters

Spillover Application	Spatial Dependence (ρ)	Spillover Effect (ψ)	Simple Random Sampling			Block Stratified Sampling		
			MSE	SD	Bias	MSE	SD	Bias
Control Only Spillover	Yes (0.01)	0.5	0.30384	0.40525	-0.26984	0.16515	0.02943	-0.34903
		0.6	0.38161	0.40623	-0.34198	0.26900	0.04620	-0.47130
		0.7	0.47105	0.41430	-0.41415	0.40359	0.06531	-0.59377
		0.8	0.57345	0.43896	-0.48647	0.56900	0.08672	-0.71637
	No (0.00)	0.5	0.17278	0.11696	-0.34278	0.26160	0.00946	-0.49594
		0.6	0.24823	0.16834	-0.41130	0.36608	0.00998	-0.59661
		0.7	0.33754	0.22914	-0.47979	0.49203	0.00951	-0.69728
		0.8	0.44075	0.29940	-0.54826	0.63946	0.00806	-0.79795
Intervention & Control Spillover	Yes (0.01)	0.5	0.09543	0.17615	0.03072	0.16515	0.02943	-0.34903
		0.6	0.11542	0.20822	0.02984	0.26900	0.04620	-0.47130
		0.7	0.13772	0.24225	0.02862	0.40359	0.06531	-0.59377
		0.8	0.16216	0.27871	0.02714	0.56900	0.08672	-0.71637
	No (0.00)	0.5	0.00790	0.04384	-0.01400	0.26160	0.00946	-0.49594
		0.6	0.01090	0.06137	-0.01679	0.36608	0.00998	-0.59661
		0.7	0.01450	0.08250	-0.01964	0.49203	0.00951	-0.69728
		0.8	0.01872	0.10723	-0.02255	0.63946	0.00806	-0.79795

Table 1: SRS vs. BSS: Intervention Effect Error Estimates - 2x4 Spatial Grid Setting

3.6 Intervention Effect Estimates (3x3 grid)

The results for the 3x3 spatial grid are summarized in Table 2 and Figure 11.

For the Control Clusters Only (TrtNoSpill) spillover application, BSS consistently yielded significantly lower MSE, SD, and bias compared to SRS. For instance, with $\rho = 0.00$, BSS MSE was minimal (e.g., 0.00069 for $\psi = 0.5$), while SRS MSE ranged from 0.09200 to 0.23485. This pattern held true with spatial dependence ($\rho = 0.01$), where BSS MSE remained very low (e.g., 0.00440 for $\psi = 0.5$), contrasting with SRS MSE values ranging from 0.25927 to 0.37154. SRS estimates consistently showed negative bias, increasing with ψ , while BSS bias was negligible.

In the Intervention & Control Clusters (TrtSpill) spillover scenario, BSS continued to demonstrate superior performance with substantially lower MSE, SD, and bias compared to SRS. For $\rho = 0.00$, BSS MSE was consistently around 0.00044, whereas SRS MSE ranged from 0.00233 to 0.00541. With $\rho = 0.01$, BSS MSE was also consistently very low (e.g., 0.03899 for $\psi = 0.5$), while SRS MSE ranged

from 0.10339 to 0.16807.

Similar to the 2×4 grid, the presence of spatial dependence ($\rho = 0.01$) generally increased MSE for both sampling methods across both spillover application types for the 3×3 grid.

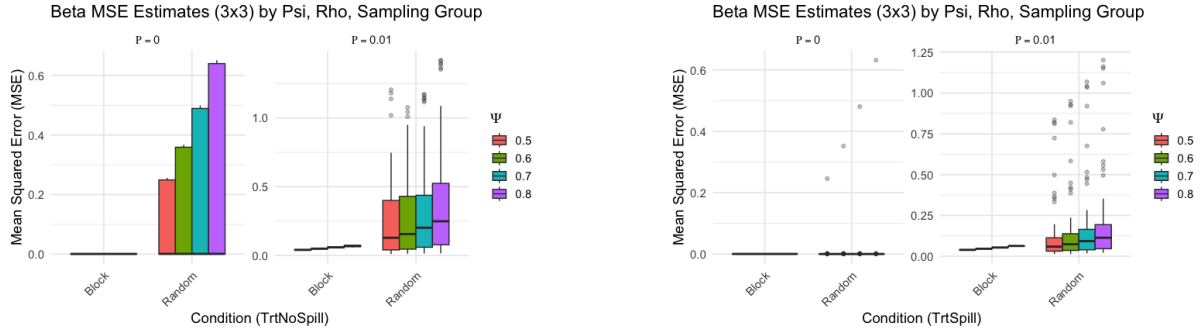


Figure 11: Intervention effect MSE for 3x3 cluster grid with spillover allowed on control and intervention clusters

Spillover Application	Spatial Dependence (ρ)	Spillover Effect (ψ)	Simple Random Sampling			Block Stratified Sampling		
			MSE	SD	Bias	MSE	SD	Bias
Control Only Spillover	Yes (0.01)	0.5	0.23379	0.25927	-0.05399	0.04054	0.00440	-0.01229
		0.6	0.27638	0.28599	-0.09229	0.04960	0.00797	-0.01773
		0.7	0.32234	0.32838	-0.13104	0.05919	0.01144	-0.02204
		0.8	0.37154	0.38566	-0.16998	0.06920	0.01446	-0.02516
	No (0.00)	0.5	0.09200	0.12071	-0.18211	0.00069	0.00020	0.00280
		0.6	0.13221	0.17397	-0.21877	0.00070	0.00018	0.00289
		0.7	0.17980	0.23701	-0.25545	0.00071	0.00017	0.00298
		0.8	0.23485	0.30992	-0.29217	0.00071	0.00017	0.00305
Intervention & Control Spillover	Yes (0.01)	0.5	0.10339	0.14801	0.07140	0.03899	0.00222	-0.01899
		0.6	0.12214	0.16878	0.08204	0.04588	0.00271	-0.02000
		0.7	0.14332	0.19051	0.09237	0.05355	0.00319	-0.02101
		0.8	0.16807	0.21492	0.10133	0.06303	0.00074	-0.01899
	No (0.00)	0.5	0.00233	0.02188	-0.00357	0.00052	0.00044	0.00383
		0.6	0.00318	0.03131	-0.00437	0.00052	0.00044	0.00383
		0.7	0.00420	0.04276	-0.00518	0.00052	0.00044	0.00383
		0.8	0.00541	0.05624	-0.00600	0.00052	0.00044	0.00383

Table 2: SRS vs. BSS: Intervention Effect Error Estimates - 3x3 Spatial Grid Setting

3.7 Intervention Effect Estimates (3x4 grid)

Table 3 and Figure 12 present the intervention effect estimates for the 3x4 spatial grid.

Under the Control Clusters Only (TrtNoSpill) spillover condition, BSS consistently outperformed SRS in terms of MSE, SD, and bias. With $\rho = 0.00$, BSS MSE was extremely low (e.g., 0.00036 for $\psi = 0.5$), while SRS MSE ranged from 0.03425 to 0.21912. This trend persisted with $\rho = 0.01$, where BSS MSE remained low (e.g., 0.03463 for $\psi = 0.5$), compared to SRS MSE ranging from 0.12009 to 0.20387. SRS estimates generally showed a small positive or negative bias, while BSS bias was consistently negligible.

For the Intervention & Control Clusters (TrtSpill) spillover application, BSS again showed superior performance with lower MSE, SD, and bias. With $\rho = 0.00$, BSS MSE was consistently around 0.00028, while SRS MSE ranged from 0.00073 to 0.02853. With $\rho = 0.01$, BSS MSE was also low (e.g., 0.22286 for $\psi = 0.5$), while SRS MSE ranged from 0.11193 to 0.19046.

Across all grid sizes and conditions, increasing the spillover effect magnitude (ψ) generally led to an increase in MSE for both sampling methods. The presence of spatial dependence ($\rho = 0.01$) consistently resulted in higher MSE for both SRS and BSS compared to scenarios without spatial dependence ($\rho = 0.00$). While SRS sometimes showed lower MSE when spillover was applied to both intervention and control clusters, BSS consistently demonstrated lower variance and often lower bias, particularly when spillover was restricted to control clusters only. This highlights that BSS provides more consistent estimates, even if SRS might occasionally yield a lower average error depending on the specific spillover dynamics.

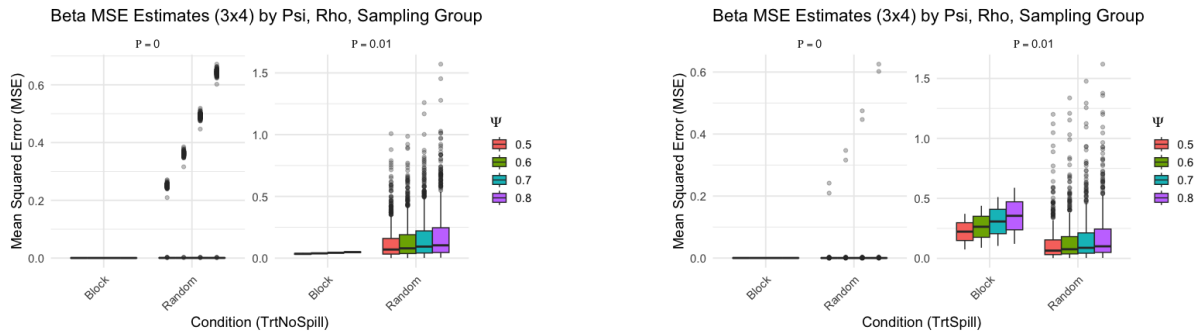


Figure 12: Intervention effect MSE for 3x4 cluster grid with spillover allowed on control and intervention clusters

Spillover Application	Spatial Dependence (ρ)	Spillover Effect (ψ)	Simple Random Sampling			Block Stratified Sampling		
			MSE	SD	Bias	MSE	SD	Bias
Control Only Spillover	Yes (0.01)	0.5	0.12009	0.13244	0.03254	0.03463	0.00963	-0.15200
		0.6	0.13800	0.14807	0.02057	0.03881	0.00505	-0.16651
		0.7	0.15902	0.17166	0.00943	0.04345	0.00011	-0.18039
		0.8	0.18330	0.20387	-0.00075	0.04842	0.00561	-0.19332
	No (0.00)	0.5	0.03425	0.08554	-0.06751	0.00036	0.00022	-0.00571
		0.6	0.04905	0.12318	-0.08097	0.00036	0.00022	-0.00563
		0.7	0.06666	0.16769	-0.09444	0.00035	0.00020	-0.00555
		0.8	0.08678	0.21912	-0.10791	0.00034	0.00016	-0.00553
Intervention & Control Spillover	Yes (0.01)	0.5	0.11193	0.12873	0.02201	0.22286	0.20995	-0.41486
		0.6	0.13069	0.14762	0.02096	0.26343	0.24750	-0.45154
		0.7	0.15112	0.16817	0.01963	0.30741	0.28774	-0.48833
		0.8	0.17323	0.19046	0.01810	0.35475	0.33060	-0.52516
	No (0.00)	0.5	0.00073	0.01051	-0.00126	0.00028	0.00004	-0.00251
		0.6	0.00096	0.01542	-0.00149	0.00029	0.00004	-0.00241
		0.7	0.00124	0.02143	-0.00172	0.00030	0.00005	-0.00232
		0.8	0.00158	0.02853	-0.00194	0.00031	0.00005	-0.00224

Table 3: SRS vs. BSS: Intervention Effect Error Estimates - 3x4 Spatial Grid Setting

3.8 Comparison of Methods

4 Application

NEED ADDITIONAL INFO ON STUDY PROPOSALS

We apply these findings to a pair of studies in development. One study is supported by staff at the Georgia Department of Community Supervision and the other is a study in development for the North Carolina Department of Adult Corrections. The investigators leading each study both seek recommendations for treatment assignment methodology.

4.1 GA Department of Community Supervision

4.1.1 Motivating Example

4.1.2 Recommendation

4.2 NC Department of Adult Corrections

4.2.1 Motivating Example

4.2.2 Recommendation

5 Discussion

5.1 Relevance of results

This simulation study provides insights into the efficiency of treatment assignment methodologies in spatial cluster randomized trials (CRTs) when considering spillover effects and spatial dependence. A

primary contribution of this work is the direct comparison between Simple Random Sampling (SRS) and Block Stratified Sampling (BSS) across various scenarios, offering practical guidance for investigators designing such trials.

Our findings show a clear relationship between the magnitude of spillover effects and the estimation error of the intervention effect. As the spillover effect (ψ) increases, the Mean Squared Error (MSE) for the intervention effect (β) generally rises, regardless of the sampling method or the presence of spatial dependence. This highlights the importance of accounting for spillover effects, as their increasing influence can diminish the accuracy of intervention effect estimates. The results indicate that randomly selected treatment assignments lead to reduced average error (lower bias) when estimating intervention effects, while block stratified treatment assignments consistently result in reduced variation (lower variance) among a series of estimates. This suggests a trade-off: SRS provides a more accurate central estimate on average, whereas BSS offers greater precision and consistency across repeated trials, potentially with higher bias.

The study also clarifies the impact of spatial dependence (ρ) on intervention effect estimation error. The presence of even a small degree of spatial dependence (e.g., $\rho = 0.01$) generally increased MSE for both sampling methods. This indicates that spatial correlation among observations can impede the accurate estimation of intervention effects if not managed during the design phase. The observed patterns across different spatial grid dimensions (2×4 , 3×3 , and 3×4) support these conclusions, demonstrating the consistency of the findings across varying spatial configurations.

The investigation into cluster spillover restrictions (spillover to control clusters only vs. spillover to both control and intervention clusters) also identified important differences. When spillover was restricted to control clusters only, the MSE for the intervention effect was higher for SRS compared to BSS, particularly with increasing spillover magnitude. Conversely, when spillover was permitted for both control and intervention clusters, SRS often exhibited lower MSE, especially when spatial dependence was absent. This understanding of spillover application types provides investigators with a more precise basis for selecting a sampling strategy that aligns with their specific hypotheses regarding how an intervention's effects might propagate.

In summary, this research offers a comparison of two sampling methodologies in spatial CRTs. The patterns observed regarding spillover magnitude, spatial dependence, and spillover application types provide practical information for practitioners. Investigators must consider the goal of an unbiased average estimate (favored by SRS) versus the need for consistent, low-variance estimates (achieved with BSS), especially when designing studies where spillover effects are expected.

5.2 Limitations

While this simulation study provides insights, it is important to acknowledge several limitations that may affect the generalizability and scope of its findings.

The study encountered issues related to multicollinearity between the direct (intervention) and indirect (spillover) effects. In some spatial configurations and with specific spillover magnitudes, the high correlation between the x_i (intervention indicator) and z_i (spillover indicator) variables in the spatial autoregressive model (Equation 2) could lead to instability in parameter estimates. While the SAR model addresses spatial relationships, extreme collinearity can still impact the precision of individual coefficient estimates, potentially affecting MSE calculations. This issue in separating highly correlated effects is known in spatial causal inference and requires careful consideration in applied settings.

The expanded simulation space quickly becomes computationally infeasible. The current study explored a specific range of spillover effect sizes, spatial dependence parameters, and grid dimensions. However, a more extensive exploration of additional parameters (e.g., different baseline effects, varying intervention effect sizes, or a wider range of spatial dependence values) or more complex spatial structures would require significantly greater computational resources. This limitation restricted the number of scenarios that could be thoroughly investigated within the current framework.

A notable limitation is the rudimentary spillover mechanism (binary effect) applied. This simulation

study modeled spillover as a fixed, binary event, meaning a unit either experienced the full potential spillover effect or none, based solely on contiguity (Rook neighbors). This approach does not account for more realistic scenarios where spillover effects may decay with distance, or where cumulative exposure from multiple treated neighbors could lead to additive or non-linear effects. The absence of more sophisticated distance-based decay functions (e.g., linear, categorical, exponential, or Gaussian decay) or mechanisms for additive spillover limits the direct applicability of these specific quantitative results to real-world situations where such nuanced spillover dynamics are likely. The current model also did not consider higher-order neighbors beyond immediate (Rook) contiguity.

The study utilized homogeneous cluster sizes and regular grid shapes. Real-world clusters often exhibit irregular shapes and varying sizes, which could influence spillover dynamics and the efficiency of different sampling strategies. The current findings may not fully extend to such heterogeneous spatial arrangements.

The simulation analysis also focused on a single outcome variable. In practical applications, interventions may affect multiple outcomes, and the optimal sampling strategy or the performance of a given strategy could differ across these outcomes. The study does not explore the implications for multi-outcome trials.

Finally, the study assumes a fixed treatment allocation ratio (primarily 1:1 or near 1:1). Varying the proportion of intervention to control clusters could impact the statistical power and bias of intervention effect estimates, especially in the presence of spillovers.

5.3 Future Considerations

Building on this study’s findings, several areas for future research can further improve our understanding of sampling design in spatial cluster randomized trials.

A key next step involves incorporating additional spillover structures beyond the binary mechanism. Future simulations should explore various distance-based decay functions, such as linear, categorical/step-wise, exponential, or Gaussian decay. This would provide a more realistic representation of how intervention effects attenuate over space and allow for the identification of optimal sampling strategies under different decay profiles. For example, an intervention with rapid exponential decay might benefit from a different sampling approach than one with a more gradual linear decay.

Furthermore, the current model’s assumption of non-additive spillover effects could be relaxed. Future research should consider the implications of “additive” spillover, where being a neighbor to multiple intervention cells allows for a greater cumulative spillover effect. This would reflect scenarios where the intensity of exposure to the intervention’s indirect effects scales with the number or proximity of treated neighbors. Understanding how sampling methods perform under such additive spillover mechanisms is important for interventions where collective exposure drives indirect effects.

Exploring other neighbor structures beyond Rook contiguity is also warranted. Investigating Queen neighbors (sharing an edge or vertex) or higher-order neighbors (neighbors of neighbors) would provide a more complete understanding of how different definitions of “proximity” influence spillover propagation and, consequently, the performance of various sampling designs.

Additionally, future work could investigate the impact of “prior” weighting of intervention based on cluster subject density. In real-world settings, clusters rarely have uniform population density. Incorporating variable subject densities within clusters and exploring how this influences spillover effects and optimal sampling strategies would add realism to the simulations.

An extension of the simulation parameters to cover a broader range of values for spillover effects, spatial dependence, and grid dimensions, potentially using advanced computational techniques or cloud computing, would allow for a more comprehensive mapping of the parameter space and a more robust set of recommendations for practitioners. This would enable a deeper understanding of the conditions under which each sampling method (SRS vs. BSS) is most advantageous, leading to more efficient and reliable designs for spatial cluster randomized trials.

Future research could also investigate the robustness of findings to model misspecification. This

includes evaluating how different sampling strategies perform when alternative spatial models (e.g., Spatial Error Model, Spatial Durbin Model) are used for analysis, or when the true underlying spatial process deviates from the assumed SAR model.

Finally, exploring optimal allocation ratios of intervention to control clusters in the presence of spillovers and spatial dependence could provide valuable insights for maximizing statistical power or minimizing estimation error. Incorporating cost-effectiveness analysis into the design framework would also be beneficial, as different sampling strategies may have varying logistical and financial implications in real-world implementation.

References

- [1] Karim Anaya-Izquierdo and Neal Alexander. Spatial regression and spillover effects in cluster randomized trials with count outcomes. *Biometrics*, 77(2):490–505, June 2021.
- [2] Luc Anselin. Spatial Econometrics. In Badi H. Baltagi, editor, *A Companion to Theoretical Econometrics*, pages 310–330. Wiley, 1 edition, January 2003.
- [3] Jade Benjamin-Chung, Jaynal Abedin, David Berger, Ashley Clark, Veronica Jimenez, Eugene Konagaya, Diana Tran, Benjamin F Arnold, Alan E Hubbard, Stephen P Luby, Edward Miguel, and John M Colford. Spillover effects on health outcomes in low- and middle-income countries: a systematic review. *International Journal of Epidemiology*, 46(4):1251–1276, August 2017.
- [4] Jade Benjamin-Chung, Haodong Li, Anna Nguyen, Gabriella Barratt Heitmann, Adam Bennett, Henry Ntuku, Lisa M. Prach, Munyaradzi Tambo, Lindsey Wu, Chris Drakeley, Roly Gosling, Davis Mumbengegwi, Immo Kleinschmidt, Jennifer L. Smith, Alan Hubbard, Mark Van Der Laan, and Michelle S. Hsiang. Extension of efficacy range for targeted malaria-elimination interventions due to spillover effects. *Nature Medicine*, 30(10):2813–2820, October 2024.
- [5] Jinxin Bian, Decai Tang, and Yan Fang. The Dual Impact and Spatial Spillover Effects of the Digital Economy on Urban–Rural Integration. *Sustainability*, 17(2):545, January 2025.
- [6] Roger Bivand. R Packages for Analyzing Spatial Data: A Comparative Case Study with Areal Data. *Geographical Analysis*, 54(3):488–518, July 2022.
- [7] Roger Bivand, Jan Hauke, and Tomasz Kossowski. Computing the Jacobian in Gaussian spatial autoregressive models: An illustrated comparison of available methods. *Geographical Analysis*, 45(2):150–179, April 2013.
- [8] Roger Bivand, Edzer J. Pebesma, and Virgilio Gómez-Rubio. *Applied Spatial Data Analysis with R*. Use R! Springer New York, New York, NY, 2008.
- [9] Roger Bivand and Gianfranco Piras. Comparing Implementations of Estimation Methods for Spatial Econometrics. *Journal of Statistical Software*, 63(18), 2015.
- [10] Roger S. Bivand and David W. S. Wong. Comparing implementations of global and local indicators of spatial association. *TEST*, 27(3):716–748, September 2018.
- [11] Michael J. Campbell and Stephen John Walters. *How to design, analyse and report cluster randomised trials in medicine and health related research*. John Wiley & Sons, Chichester, West Sussex, 2014.
- [12] Allan Donner and Neil Klar. *Design and analysis of cluster randomization trials in health research*. Arnold, London, 1. publ edition, 2000.
- [13] A. Stewart Fotheringham and Peter Rogerson. *The Sage handbook of spatial analysis*. The Sage handbook of. Sage, London, 2009.

- [14] R. C. Geary. The Contiguity Ratio and Statistical Mapping. *The Incorporated Statistician*, 5(3):115, November 1954.
- [15] Richard J. Hayes and Lawrence H. Moulton. *Cluster Randomised Trials, Second Edition*. Chapman and Hall/CRC, July 2017.
- [16] K Hemming, S Eldridge, G Forbes, C Weijer, and M Taljaard. How to design efficient cluster randomised trials. *BMJ*, page j3064, July 2017.
- [17] Michael G Hudgens and M. Elizabeth Halloran. Toward Causal Inference With Interference. *Journal of the American Statistical Association*, 103(482):832–842, June 2008.
- [18] C Jarvis. Spatial Analysis of Cluster Randomised Trials. 2018. Publisher: London School of Hygiene & Tropical Medicine.
- [19] Christopher Jarvis, Gian Luca Di Tanna, Daniel Lewis, Neal Alexander, and W. John Edmunds. Spatial analysis of cluster randomised trials: a systematic review of analysis methods. *Emerging Themes in Epidemiology*, 14(1):12, September 2017.
- [20] John N.S. Matthews. *Introduction to Randomized Controlled Clinical Trials*. Chapman and Hall/CRC, 0 edition, June 2006.
- [21] Paula Moraga. *Spatial statistics for data science: theory and practice with R*. Chapman and Hall/CRC Data Science Series. CRC Press, Boca Raton London New York, first edition edition, 2024.
- [22] P. A. P. Moran. The Interpretation of Statistical Maps. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 10(2):243–251, July 1948.
- [23] P. A. P. Moran. Notes on Continuous Stochastic Phenomena. *Biometrika*, 37(1/2):17, June 1950.
- [24] David M. Murray. *Design and analysis of group-randomized trials*. Number v. 27 [i.e. 29] in Monographs in epidemiology and biostatistics. Oxford University Press, New York, 1998.
- [25] Georgia Ntani, Hazel Inskip, Clive Osmond, and David Coggon. Consequences of ignoring clustering in linear regression. *BMC Medical Research Methodology*, 21(1):139, December 2021.
- [26] Keith Ord. Estimation Methods for Models of Spatial Interaction. *Journal of the American Statistical Association*, 70(349):120–126, March 1975.
- [27] Edzer Pebesma. Simple Features for R: Standardized Support for Spatial Vector Data. *The R Journal*, 10(1):439, 2018.
- [28] Edzer Pebesma and Roger Bivand. *Spatial Data Science: With Applications in R*. Chapman and Hall/CRC, New York, 1 edition, May 2023.
- [29] W. R. Tobler. A Computer Movie Simulating Urban Growth in the Detroit Region. *Economic Geography*, 46:234, June 1970.
- [30] Aman Ullah and David E. A. Giles. *Handbook of Applied Economic Statistics*. Number v. Vol. 155 in Statistics: a Series of Textbooks and Monographs. Chapman and Hall/CRC, Boca Raton, 1998.
- [31] Lance A. Waller. *Applied Spatial Statistics for Public Health Data*. Number v.368 in Wiley Series in Probability and Statistics Ser. John Wiley & Sons, Incorporated, Hoboken, 1st ed edition, 2004.